HAT-P-54b

R-1606

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-54_190112` · created 2026-08-02T02:53:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458495.6206 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.047
Transit depth (Rp/Rs)^2	2.18 +/- 1.39 [%]
Orbital Inclination (inc)	89.8 +/- 2.22
Airmass coefficient 1 (a1)	1.063 +/- 0.023
Airmass coefficient 2 (a2)	-0.041 +/- 0.017
Scatter in the residuals of the lightcurve fit is	2.16 %
Best Comparison Star	#1 - [305, 39]
Optimal Aperture	2.94
Optimal Annulus	12.06
Transit Duration (day)	0.0651 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.047 vs 0.1572 (deviation 0.19σ)
Duration fit vs published	0.0651 d vs 0.075 d (deviation 2.38σ)
Mid-time O–C	0.9 min
Depth SNR	2.8
Residual scatter	2.16 %

Light curve

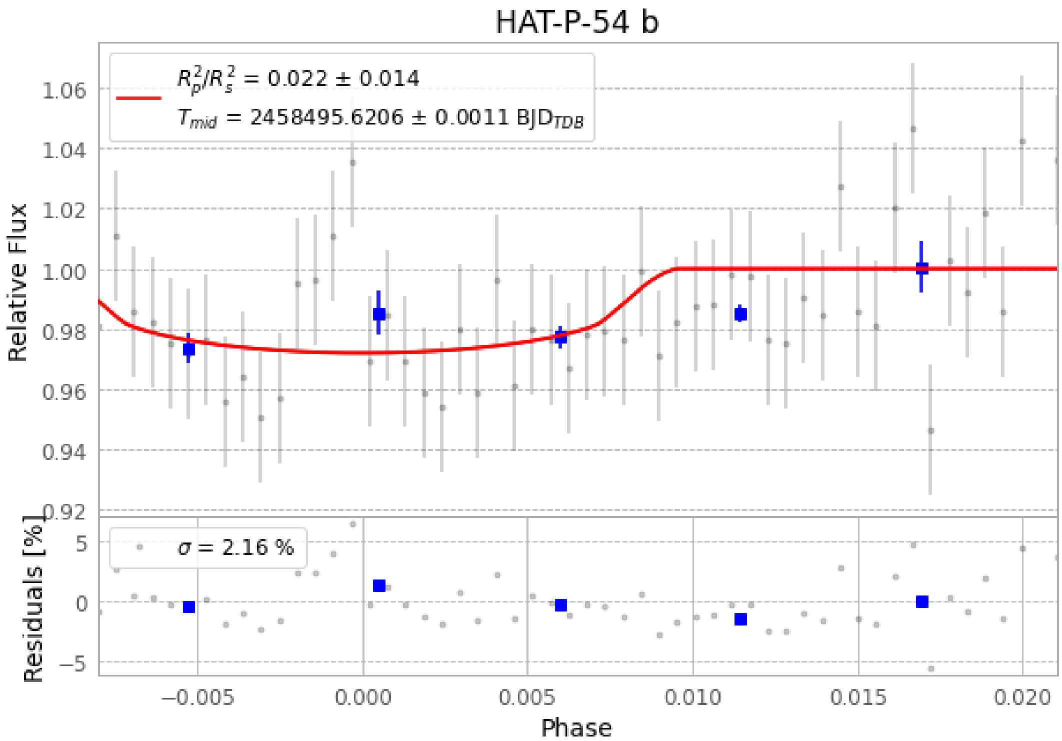

Figure 1 — FinalLightCurve_HAT-P-54 b_2019-01-11.png (SHA-256 3d87332fd050dcbc…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 150], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 57a99783dc4fa4cf…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

54 FITS frames (35 MB), `HATP-54190112020012.FITS` → `HATP-54190112043912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-190112.log (6e1fa5921324…), FinalLightCurve_HAT-P-54 b_2019-01-11.png (3d87332fd050…), FinalLightCurve_HAT-P-54 b_2019-01-11.csv (892c78ec329d…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)